Multi-Market Contact in International Trade; Evidence from U.S. Battery Exporters

by

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Abstract

When competitors compete in more than one market they are said to have multi-market contact (MMC). Firms with MMC are more likely collude to avoid cross-market retaliation. This paper investigates the impact of MMC among U.S. battery exporters on the prices they set in foreign markets using confidential export transaction data provided by the U.S. Census Bureau. The ability of firms to exploit MMC for collusive gain in international markets can be both detrimental to import-dependent consumers and harder for anti-trust authorities to detect. Motivated by litigation finding evidence of collusive behavior by multi-national battery manufacturers, MMC has an upward effect on export prices set by U.S. battery exporters. These results are robust across different panel regression specifications using different measures of MMC.

Keyword: multi-market contact, oligopoly, export prices, collusion

JEL Classification: D43, F12, F13, F14, L13, L14, L40, L63

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I am an independent researcher with interests in international trade and applied game theory. I currently work as an economist for a U.S. federal agency. I received my PhD in economics from American University with a focus in international trade and industrial organization. I wrote this paper in my personal capacity. All thoughts expressed in this paper are my own and do not reflect those of the U.S. government. I was a part-time Census Bureau employee and a graduate student at American University when I started this research.
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1 Introduction

In 2013, battery purchasers accused several lithium-ion battery (LIB) producers of engaging in a price-fixing cartel in the United States and elsewhere.[1] The first complaint was filed in late 2012 and the case took five years to come to a verdict. The plaintiffs alleged that manufacturers secretly met to coordinate prices and allocate markets. Most of the defendants pled guilty and paid the plaintiffs millions of dollars in damages (U.S. Department of Justice (2010)).

Most interestingly, these manufacturers were able to enforce adherence to the collusive agreements in place by agreeing to price batteries for computer laptops at predetermined levels and shared information to enforce adherence to maintain artificially high prices (U.S. Department of Justice (2010)). Motivated by a different collusion case involving multinational vitamin manufacturers, Choi and Gerlach (2012) theorize that cartels are more likely to succeed if members can coordinate pricing strategies across international markets. Zhou (2016) finds empirical support for this theory by first observing that a majority of cartel members prosecuted by the European Comission (EC) met in multiple markets and then finding that anti-trust regulation in one market decreased the probability of joining the cartel in different markets. Understanding the antecedents of collusive behavior in international markets is an important step to preventing global cartels from exploiting consumers.

When the same set of firms competes in multiple markets, they are said to have multimarket contact (MMC).[2] One result of MMC could be an increased willingness to operate at a collusive equilibrium in order to avoid cross market retaliation. Edwards (1955) first posited this theory of "mutual forbearance" (MF). This paper explores the effect of MMC on firm export pricing strategies using transaction level trade data collected by the U.S. Census Bureau. I focus on U.S. battery exporters where collusive behavior by certain types of battery

[1] See https://www.justice.gov/opa/pr/panasonic-and-its-subsidiary-sanyo-agree-plead-guilty-separate-price-fixing-conspiracies for details of the case brought against these suppliers.

[2] Examples of this include two airlines offering trips on two or more routes, two cement firms owning production facilities in different regional markets, or a single set of competitors exporting a product to the same countries.

producers (e.g. LIB manufacturers) has been well documented. Using conclusions from a EC analysis on identifying appropriate battery markets, I define markets as country-HS pairs (Council of European Union (2009)).[3]

A mutual forbearance outcome can arise from MMC by formalizing two-firm/two-market oligopoly competition models where the optimal response for a competitor would result in actions taken in multiple markets (Bernheim and Whinston (1990), Thomas and Willig (2006)). The general framework asserts that firms may be wary of competing aggressively against competitors they face in many markets to avoid potentially complex punishment schemes after deviating from the cooperative equilibrium. For instance, if two firms compete in two markets, then a deviation from the collusive strategy could lead to retaliation in both markets. If those markets are asymmetric then it is more likely that colluding will reap more lifetime profit than a short-term gain from undercutting their competitor.[4]

The effects of MMC have been most studied in the empirical industrial organization literature.[5] Most work corroborates the MF hypothesis, but few address MMC outside of nationally defined markets. One exception is Tieying et al. (2009), who finds that MMC decreases the likelihood of competitive action by firms in individual export markets.[6] Additionally, Alcantara and Mitsuhashi (2015) show that MMC among Japanese auto-parts exporters tend to dampens these firms' FDI in foreign countries. Finally, Feinberg (2015) uses bilateral country and product level U.N. Comtrade (UNCTD) data to show that MMC among prominent exporting countries in the fats and oils trade has a significant and positive relationship on country level export prices.

While these papers have suggested that mutual forbearance arising from MMC potentially exists in international trade, they have not definitively explored the result using firm-level

[3]HS stands for "Harmonized System" and is used by governments to classify products in customs records.

[4]Bernheim and Whinston (1990) shows that 2 firms competing in identical markets replicates the single market outcome.

[5]For some notable studies that look at the impact of MMC on price, see Feinberg (1985), Evans and Kessides (1994), Jans and Rosenbaum (1997), Parker and Röller (1997), Feinberg (2014) and Ciliberto and Williams (2014). See Yu and Cannella (2013) for a survey of the literature.

[6]In their paper, competitive action was determined through text analysis of automotive news articles.

price setting strategies in foreign markets. In addition, I implement different measures of MMC to assess the robustness of the relationship. Overall, I find that firms charge higher prices in markets where they have MMC. More specifically, a one standard deviation increase in a firm's MMC in a market corresponds to roughly a 1% increase in the price it sets in that market. These results are consistent with those highlighted in other empirical studies (Ciliberto and Williams (2014), Feinberg (2015), Evans and Kessides (1994)). Finally, I show that this result is robust across different econometric specifications, definitions of markets and underlying sample of likely competitors. Overall, my findings provide further evidence of collusive behavior arising from MMC.

Section 2 introduces the theory behind how MMC could lead to an increased likelihood of collusion. Section 3 motivates the susceptibility of the battery industry to MF from MMC (*Hanlon vs. LG Chem* (2017), Council of European Union (2009)) using market research on the U.S. battery industry and export activity. Section 4 introduces the data and describes steps taken to identify the correct sample of battery competitors and define market boundaries. Section 5 describes the empirical specification and addresses estimation issues related to the effects of external competition stemming from foreign countries. Section 6 shows the main results in addition to some key robustness checks. Finally, Section 7 concludes with some suggestions for future research.

2 Theory

Bernheim and Whinston (1990) finds that two firms competing within a single market will play the Nash equilibrium strategy or behave competitively. When firms compete across multiple markets, however, the strategic interaction between firms becomes more complex. On the right side of Figure 1 is a diagrammatic representation of 2 firms competing with each other across 2 markets, but each firm having a "sphere of influence" in an opposing market.[7] In this case, firm A has a bigger presence in market 1 and firm 2 has a bigger presence in

[7] An example of sphere of influence would be firm A having lower costs in Market 1 than firm B.

market B. If firm A wanted to take market share from firm B in market 2 then it would need to consider that firm 2 could retaliate in market 2, market 1 or both. This can leave both firms hesitant to take competitive action against each other for fear of losing ground in their more valuable market. Hence a mutual forbearance result arises. In the appendix, I discuss this theory formulated by Bernheim and Whinston (1990) more formally from the perspective of an iterated Prisoner's Dilemma game. Unfortunately, to my knowledge, there are no theoretical models that look at more than $M > 2$ firms competing across $N > 2$ markets. However, many notable studies in this area have found empirically studied M firms competing in N markets (See Evans and Kessides (1994), Jans and Rosenbaum (1997) for notable examples) and found that MMC does correlate significantly with less competitive outcomes.

3 The Global Battery Industry

In the average year from 1997-2013, Tables 4 and 5 indicate there are roughly 350 manufacturing firms that export batteries and around 1,875 non-manufacturing firms export batteries, respectively.[8] According to publicly available information from the Economic Census, there are roughly 200 establishments that have battery production as their primary activity in 2012. Original equipment manufacturers (OEMs), for instance manufacturers of computer and automobile products, are typical consumers for battery suppliers (Hoover's (2017)). However, these same OEMs can also turn against their battery suppliers and behave competitively in the battery market (IBISWorld (2017)). For instance, Daimler, a German car manufacturer, opened 5 battery manufacturing plants in Asia, Europe and North America.[9] Non-manufacturing exporters are typically wholesalers or retailers that largely participate

[8] I define manufacturers as a firm owning at least one manufacturing establishment. See Section 4 for more information.

[9] See https://media.daimler.com/marsMediaSite/en/instance/ko/Under-the-microscope-Battery-production-Daimler-to-build-global-production-compound-for-batteries.xhtml?oid=29798176.

in battery aftermarkets.[10] It is common for even medium-sized battery suppliers to own subsidiaries abroad (IBISWorld (2017)), meaning that many foreign owned battery firms are likely to have subsidiaries exporting from the U.S.

Figure 2 shows U.S. export trends for rechargeable and non-rechargeable batteries from 1992 - 2016 from the UN Comtrade Database. Interestingly, In 2000, around the same time collusive activities allegedly started between battery suppliers, there was a precipitous drop in U.S. battery exports. U.S. exports in non-rechargeable batteries are small in comparison and have remained relatively steady from 1992-2016. This is consistent with market research indicating that battery manufacturers are more focusing on non-rechargeable battery production (IBISWorld (2017)). Furthermore, according to Figure 3, U.S. battery exports reach a wide range of different countries from year-to-year.

Previous empirical studies have not investigated the effects of MMC in the battery industry, let alone in an international trade context. Feinberg (2018) identifies the prerequisite industry characteristics likely to engender MF from MMC. These are:

- Clear market boundary identification.
- Oligopolistic and concentrated markets.
- Quasi-centralized decision making.

Any study using measures of MMC needs to clearly discern between different markets. Defining markets too narrowly may result in omitting some competitors from the computation, leading to a possible downward bias in measurement of MMC, but it may also lead to overestimating the MMC measure by counting existing rivals more frequently across markets. The opposite is true for overly broad market boundaries. Although the direction of bias is an open question with arbitrarily drawn boundaries, preliminary research on alternative industry market definitions should be done to mitigate spurious relationships in the data

[10] An aftermarket is a market for spare or unused parts. In this case, one example of an aftermarket would be the sale of batteries a manufacturer was unable to sell itself.

(Feinberg (2018)). Using the rich data structure of trade data introduced in Section 4, I define markets as country-HS6 pairs.[11]

The battery industry differs from these prior MMC studies in that battery types, in general, are substantially differentiated. However, a regulatory decision by the European Commission with regard to a proposed merger by Panasonic and Sanyo, finds that the battery industry is also highly segmented across battery types. Figure 4 shows how the EC defines product markets within the battery industry. It asserts that product markets are clearly divided by different chemistries (i.e. at the HS6 level), but product markets for some non-rechargeable and rechargeable LIB batteries can be further subdivided by shape and sub-chemistry (Council of European Union (2009)).[12] Furthermore, these more specialized batteries are often tailored according to custom specifications that are negotiated between the buyer (e.g. OEM) and supplier during the contract negotiation process (Hoover's (2017)). Hence competition, or the lack thereof, between suppliers could manifest in how aggressively they negotiate a contract with buyers before production even takes place.[13]

Both the theoretical and empirical literature of MMC in international trade settings assume markets divided at the country level. For instance, Bond and Syropoulos (2008) and Choi and Gerlach (2012) assume a home and foreign country firm and Feinberg (2015) and Feinberg (2013) assume markets are defined at the country-product level. Additionally, in the Consolidated Class Action Complaint (CCAC), plaintiffs allege that firms had "conversations and communications" on "prices for [LIBs] in the *United States and elsewhere*", thus showing that manufacturers at least considered markets within the United States as distinct from international markets, but it is unclear how conspirators classified other regions (*Hanlon vs. LG Chem* (2013), p.10). It is noteworthy that there is vast evidence in the trade literature of considerable export price differences within firm-product, but across different countries.[14]

[11] In Chapter 1 of my dissertation, I conduct robustness exercises using broader definitions of markets (e.g. at the country-HS4 level).

[12] Their justification is that it would be costly for purchasers to substitute between different battery types.

[13] That being said, the possibility of quality upgrading as a response to MMC is an interesting alternative theory. I discuss this further in Section 5.

[14] See Görg et al. (2010), Manova and Zhang (2012) and Harrigan et al. (2015).

Firms with market power in multiple markets create opportunities for competitors in those markets to exercise leverage through retaliation on multiple platforms. Hence, one prerequisite for mutual forbearance to occur is that markets are concentrated and that retaliation between competing firms is possible (Ma (1998)). In the anti-trust case discussed above, manufacturers engaged in horizontal price fixing on a global scale.[15] The defendants engaged in meetings, conversations and communications in the U.S. and elsewhere for the purpose of monitoring and enforcing adherence to the agreed upon price-fixing scheme (*Hanlon vs. LG Chem* (2017)). In a related case the EU commission also found that these battery producers were engaging various markets across Europe and Asia.[16]

In international trade, many contemporary empirical models of firm-level export behavior are inspired by monopolistic competition models like Melitz (2003), where products exported by a firm are sufficiently different from those of other firms to the point where strategic interaction would not play a prominent role. However, theoretical models of MF arising from MMC stem from well-known oligopoly models such as Cournot (See Feinberg (1984)) and Bertrand (See Bernheim and Whinston (1990)). Empirical literature in MMC typically focuses on service related industries (e.g. airlines or health administration) or the sale of homogenous goods across different geographically defined markets. Ciliberto and Williams (2014) and Evans and Kessides (1994) study MMC among airline carriers, Lin and McCarthy (2018) is the latest study looking at the effects of MMC between health insurance providers and Jans and Rosenbaum (1997) focus on the ready-to-mix cement industry and delineates markets by city area in the United States.

Finally, decision making by firms should consist of some mix between headquarter and local market offices. If all the decisions are made at the headquarter level, then the notion of multiple markets is blurred and if decisions are made completely at the local level then coordination across markets is unlikely (Feinberg (2018)). For instance, a CEO might see

[15]Horizontal price fixing occurs between competitors whereas vertical price fixing would perhaps involve a subsidiary and a parent.

[16]See https://ec.europa.eu/commission/presscorner/api/files/document/print/en/ip_16_4356/IP_16_4356_EN.pdf.

one global market, but a plant manager will only see their own local market. In a class action complaint that precipitated the anti-trust verdict by the DOJ, plaintiffs alleged that "high-level executives engaged in a series of collusive meetings and communications" with the objective of "inflating [LIB] battery prices. Furthermore, "high-level executives with pricing authority discussed confidential future plans and strategies" and "discussions were also used to police, enforce and verify" that each firm was "adhering to Defendants' plan to artificially raise [LIB] prices". The qualifying language by the plaintiffs on the colluders suggest that decision making on prices and quantities were determined by company officials who were not subsidiary managers, but also not CEOs.[17].

4 Data

To study competition in multiple markets, the ideal sample would consist of all firms that strategically interact across clearly defined markets. Using stylized facts from the LIB industry (see Section 3), these would be battery manufacturers competing across country-product markets. I use a combination of product information in the Census Bureau's confidential transaction level Longitudinal Firm Trade Transaction Database (LFTTD) and establishment level information in the Longitudinal Business Database (LBD) to identify the sample and define markets. In addition, I link publicly available information from the bilateral-product level UN Comtrade Database (UNCTD) and the World Bank's Development Indicators (WDI) for information on local market characteristics such as market size and extent of local competition. Specific to international trade, I use information on related party trade to isolate buyer-seller transactions (inter-firm trade) instead of transactions between subsidiaries (intra-firm trade).[18]

My main sample is battery exporters that own a manufacturing establishment. I first identify battery exporters using HS code information in the LFTTD and then subset those

[17]See https://www.reversethecharge.com/pdf/Amended-Class-Action-Complaint.pdf

[18]In export data, related parties are defined as one party owning at least 10% stake of the other.

firms that own at least one establishment whose primary activity falls under a manufacturing NAICS code.[19] While choosing exporters that own a battery manufacturing establishment is feasible, it could also exclude potential competitors for two reasons. The first is that the Census Bureau uses the establishment's *primary* activity (defined as the activity that generates the most revenue) to assign NAICS codes. Therefore, it is certainly possible that firms could produce batteries as a secondary activity and act as a minor competitor. That being said, I do check the robustness of my key results by running the model on different sets of competitors: the set of all battery exporters and the set of battery exporters that own a battery manufacturing establishment (i.e. firms that own an establishment with the NAICS code of "33591").

The LFTTD is a database built and maintained within the U.S. Census Bureau.[20] The database contains information on the universe of transactions that make up U.S. international trade. The export data is generated by export declaration form 7525-V submitted by exporting establishments and is unique at the transaction level. Each transaction contains information on the type of product being shipped (up to 10 digit HS codes), the amount and value of each shipment, whether the transaction was between related parties (subsidiaries within a multinational, e.g.) and to where the product is being shipped (at the port level).[21] The data used in this chapter is aggregated up to the year-firm-country-HS6 level and spans the years from 1997 to 2013.[22]To emphasize transactions between buyers and sellers, I isolate arm-length transactions.[23] Battery shipments are identified by HS codes starting with "8506" (non-rechargeable fuel-cells) or "8507" (rechargeable fuel-cells) in the LFTTD. Finally, it is unlikely that the effect of MMC with firms earning minimal revenues in markets will have much effect on a larger players' actions. For this reason, I drop firms from markets

[19]Manufacturing NAICS codes are "31", "32" or "33".

[20]See Kamal and Ouyang (2020) for details on the construction of the database.

[21]HS codes refer to Schedule B numbers that U.S. exporters use to identify types of products that are exported.

[22]I look at country-HS4 markets as a robustness check.

[23]Related party transactions are shipments that go between subsidiaries within a firm. Recent research has shown that multi-nationals use transfer pricing to purposely under-price related party shipments from high-tax countries to low-tax countries to maximize profits (Cristea and Nguyen (2016), Flaaen (2017)).

if they earn less than $\$10,000$ before computing the MMC measure.

The LBD is generated from the Census Bureau's Business Register (BR) which contains all business entities in the United States. The BR is used by the Census Bureau and other statistical agencies to identify business establishments for various censuses (such as the economic census) and surveys (such as the annual survey of manufacturers).[24] The data contains establishment-level payroll and employment information. This data can be aggregated to a firm level identifier that is readily linked to other Census data. Furthermore, one can capture activity of foreign owned if they have subsidiaries operating in the United States.[25]

The UNCTD contains export and import information at the reporter, partner and six-digit HS code level from 1988 to 2017. Over 170 countries provide the UN with annual trade statistics at the commodity and country level and covers roughly 90% of total trade at the bilateral country and six-digit HS code level. This data can provide information on the level of foreign party trade (countries other than the U.S.) into every market and volume of exports from domestic competition. I am primarily concerned with retaining the structural information (importing country, exporting country, product) along with the total value of imports for each reporting country from all partner countries in HS codes starting with "8506" and "8507". Reporter and partner countries are represented by three-digit country ISO codes. I use this information to assess the extent of missed contacts coming from third-party countries.

5 Model

Equation 1 shows the baseline specification.

$$Log(P_{imt}) = \beta_{mmc}\mathbf{MMC}_{imt} + \beta_F\mathbf{F}_{it} + \beta_{FM}\mathbf{FM}_{imt} + \alpha_{im} + \gamma_t + \varepsilon_{imt} \tag{1}$$

In this analysis, P_{imt} is the average unit value that firm i sells in market m in year t

[24] In Census data, a firm is defined as a set of establishments with common ownership.
[25] Reference Chow et al. (2021) for a more detailed description of the LBD.

and is created by dividing total value of exports by number of units exported for i in m in year t.[26] I then sum total export value and number of units sold from all the exporter's arm-length HS6 transactions to the destination country made before computing unit prices. $\mathbf{MMC}_{imt}$ is a vector of MMC measures for i in m at t. $\mathbf{F}_{it}$ is a vector of firm level controls and $\mathbf{FM}_{imt}$ captures export activity of i in m. α_{mt} are market$\times$ year fixed effects. ε_{imt} are errors clustered at the market-year level.

5.1 MMC

MMC, defined at the firm-market level, is equal to the count of how many times a firm meets its rivals in other markets outside of the focal market. Figure 5 shows an example of the calculation for MMC in the data. Here, Firm 1 and Firm 2 compete in Markets 1, 2 and 3 and Firm 3 only competes in Market 1 and 2. Firms 2 and 3 are Firm 1's competitors in Market 1. Firm 1 faces Firm 2 twice in other markets and Firm 3 once in another market outside of Market 1. Hence the degree of MMC for firm 1 in market 1 is $MMC_{11} = 3$.

Generally, to compute MMC for a firm i in market m, I want to count the number of times a firm i exports to the same markets as fellow exporters also exporting to market m and subtract that number from the number of competitors in the market m. This measures the extent exporters meet in other markets *outside of market m*. Let ω_{ij} be the number of markets where firm i and j are both exporters. Equation 5.1 shows how to calculate the degree of MMC for firm i in market m.

$$MMC_{im} = \sum_{j \in C_{im}} \omega_{ij} - |C_{im}|. \tag{2}$$

C_{im} is the set of competitors for i in m and ω_{ij} is the number of export markets where competitor $j \in C_{im}$ and i both export. Hence, MMC_{im} is the number of markets i faces all

[26]Some products only have weight information (e.g. HS code 850690 and 850790, or battery parts for primary and secondary batteries, respectively) and are clearly flagged and omitted from the analysis. However, I include those markets in computing MMC.

competitors j in m in markets besides m.[27]

Table 6 shows summary statistics for MMC in addition to the number of other U.S. exporters a focal exporter is expected to meet. A firm i in a country-HS6 market m at time t can expect to meet its competitors 35 times in other markets.[28] This is despite the observation that a firm in a country-HS6 market can expect to see only 2 to 3 fellow U.S. competitors. This suggests that exporters on average are likely quite familiar with the few exporters they meet. In country-HS4 markets, MMC increases dramatically to just over 100 contacts in other markets. This implies that MMC is highly sensitive to the inherent market structure of global battery markets and that robustness checks using differing definitions of markets is necessary. Additionally, there is considerable variation in all measures of MMC used in this analysis, where a one standard deviation increase translates to at least a doubling in the frequency firms meet their competitors in other market.

5.2 Dependent Variable

Table 7 shows considerable variation in firm-market prices. After reaching a similar conclusion using the same data, Harrigan et al. (2015) finds that this variation is heavily explained by destination country, product and firm information. It will therefore be prudent to control for known effects of export price fluctuations at both the market and firm level to correctly identify effects from MMC. From the data, F-statistics output from performing one-way ANOVA tests on U.S. export prices across all country-HS6 markets for each year from 1997 to 2013 were greater than or equal to 5.2, which suggests that export prices are significantly different across these different market boundaries. Furthermore, a log-linear transformation is warranted as the distribution for export prices is heavily skewed. It is worth noting that some types of batteries can be quite expensive. For instance, the average cost of production

[27]By construction all exporters in market m will have at least one contact. Because I want to emphasize outside contacts, I want to omit the contacts firm i has with competitors in the focal market m. To do this, I subtract by the number of competitors in market m omits the number of contacts for a firm i in market m .

[28]For example, assume 2 firms compete in 3 markets. In any one market, that firm will meet it's competitors twice in other markets.

of one lithium battery can exceed $\$1,000$.[29]

5.3 Controls

Table 4 also describes firm-level controls used in the analysis. Recall that a firm is included in the sample if it ships batteries to any market *and* also own a manufacturing establishment. Notice that only 20% of electronic sales come from batteries, which implies that a majority of the sample are multi-product exporters. As mentioned in Section 4, manufacturing plants can produce multiple products and downstream OEM buyers can also themselves turn against their suppliers. Hence, it would be prudent to include these potentially minor exporters in the sample, but also control for the fact that they could be relatively small players. As a robustness check, I interact "% Total Exp. in Batt" with MMC to see how more predominant battery exporters respond to MMC compared to firms with a more diversified export portfolio.[30] Bernard et al. (2007) find that exporters tend to be larger in terms of sales and employment and more capital intensive. In addition, they find that larger exporters tend to ship more products to more destinations and experience lower trade costs. Hence, larger, more capital intensive firms will likely correlate positively with MMC and negatively with price. Table 4 shows these exporters are relatively large and tend to operate in many export markets. This means that many, instead of just a few outlying firms could coordinate actions (or resources) across different export markets.

Firm-market independent variables in the specification are meant to control for the presence of the exporter in the market. These include: firm experience in exporting into that market (measured in years) and the share of shipments going to a related party.[31] Alvarez et al. (2013) finds, using Chilean exporters, that previous experience exporting a product to a country lowers export costs related to search. Helpman et al. (2004) also finds that the

[29] See https://www.ucsusa.org/clean-vehicles/electric-vehicles/electric-cars-battery-life-materials-cost#.W-8RzOhKiUk.

[30] I run robustness checks on the set of only those battery exporters with a battery manufacturing establishment (NAICS5 = 33591).

[31] Firm experience is calculated as the number of total prior years a firm has exported a product to a country.

most cost efficient firms tend to have FDI (or related party trade). Specific to the battery industry, a subsidiary presence likely indicates that the exporter has easier (cheaper) access to buyers, which makes the firm more competitive. Therefore, the effect on intra-firm shipments should also be negative on inter-firm export prices. In Table 8, 11% of firm export transactions go between related parties or between subsidiaries owned by the firm. Furthermore, exporters are experienced in the markets. This easier access significantly lowers firms search costs, which could translate to lower prices, but it could also give competitors easier access to each other in foreign markets where firms can coordinate. The resulting effect of these effects is ambiguous, but should also be controlled for in the specification.

Since I have a wealth of information about the firm and the market on a year-to-year basis, my main specification includes market-firm fixed effects (α_{mi}) and year-fixed effects (γ_t). Including market-firm fixed effects allows us to control for confounding time-invariant firm and market factors that could effect firm pricing behavior in the market that I cannot capture in the data available. Not only do these fixed effects control for variables that include distance to the market, the political and economic environment of the country, the type of battery and various characteristics of the firm, but they also control for hard-to-capture characteristics that would effect the ability of firm i to operate in that specific market m. In addition, I also control for year-fixed effects to account for time varying factors that could effect overall U.S. export activity. Examples of these types of variables include inflation, global supply chain constraints affecting U.S. battery exporters and global battery demand.

Finally, ε_{imt} are error terms. By construction, the assertion that cross-dependencies exist across markets leads to a violation of the independence assumption for a linear regression model. Empirically, MMC_{imt} is necessarily positively correlated with MMC_{jmt}, MMC_{ilt} and MMC_{jlt} for firms i, j and markets l, m. While estimated coefficients are still consistent, the dependence between different observations of MMC can lead to over-restricted coefficient variances resulting to over-inference or mistaking significant for insignificant relationships. I mitigate this by showing results with year-firm or year-market clustered errors

in all regression specifications. Nonetheless coefficient variances will be restricted because correlation between either MMC_{imt} and either MMC_{jmt} or MMC_{ilt} and MMC_{jlt} will be unaccounted. As a key robustness check, I also use a feasible GLS estimator using random-effects estimation to check that the correlation of residuals across different clusters does not impact inference. To find exact coefficient variances, one could try using techniques from the spatial econometrics literature (I push this to future research), but the simple models presented below are sufficient to show the general relationship between the independent and dependent variables.

5.4 Estimation Issues

5.4.1 Omitted Variable Bias

In addition to firm × market and year fixed effects, I implement a series of other fixed effect specifications to assess the robustness of the main result. These include: year, year and firm, year × firm, and year, firm, country and product fixed effects. With 5,500 firm-years and 8,000 market-years, I do not have enough power in my analysis to run the full set of fixed effects. As such, I include a vector of market level controls in each robustness specification to control for the remaining variation not accounted for by the fixed effects being used.

Most notably, the level of market competitiveness in general, but specifically market concentration, are included in most other empirical MMC analyses, however they are not included in this analysis. As discussed in Section 3, I assume foreign battery markets are at least as concentrated as the moderately concentrated U.S. market since the U.S. is the biggest market for batteries and therefore likely the most competitive. Nonetheless, the differing degrees of market concentration across different markets could effect the result as shown in Feinberg (1985), Bernheim and Whinston (1990) from the economics literature and Ma (1998) and Jayachandran and Varadarajan (1999) from the strategic management literature. These papers assert that concentrated markets, indicative of high levels of market power, are a necessary prerequisite for firms to use MMC to mutually forbear. If one has earnings

information for every firm in every market, one could use a measure of market concentration to assess whether firms in the market are behaving competitively. Unfortunately, in studying this question in a global trade context, I do not have information on every firm in every market because we do not have transaction data of foreign owned firms. It is therefore especially prudent to consider market-year fixed effects as a primary robustness check to gauge whether there should be a concern about omitting market concentration from the analysis.

5.4.2 Differing Degrees of Competition

It is certainly conceivable that firms can place asymmetric importance onto different markets, meaning that they could place different weights onto each market contact it has with competitors. It is therefore appropriate to create a weighted measure of MMC inspired by (Feinberg (2015) and Feinberg (2013)), called Exports-At-Risk (XAR), which gauges the importance of outside markets relative to market m. To create XAR, one can replicate the procedure to create MMC by substituting ω_{ij} with $\hat{\omega}_{ij} = \sum_{\hat{m} \in M_i} R_{i\hat{m}} \mathbb{1}[j \in \hat{m}]$ or the total revenue firm i earns in all markets it is present (M_i) where competitor j is also present. Equation 3 shows how I calculate XAR_{im}:

$$XAR_{im} = \sum_{j \in m} \hat{w}_{ij} - |C_{im}| R_{im} \tag{3}$$

5.4.3 Sampling Bias: Isolating the Set of Competitors

One concern with using U.S. export data is that we are missing business activity from prominent producers in other countries. Especially in the battery industry, where major battery producing firms may be headquartered in other countries. However, as mentioned in Section 3, it is likely that many major foreign producers of batteries have manufacturing operations in the United States. For instance, while LG Chem is headquartered in South

Korea, they own manufacturing plants in the United States.[32] While I may not observe all of business activity of each global multinational firm in this sector, I make the arguably weak assumption that I am observing some activity from most firms in this industry.

I use the sample of all battery exporters that own a manufacturing establishment. Admittedly, this sample is a bit broader than the set of firms that self define primarily as battery manufacturers. However, as mentioned in Section 3, manufacturing establishments, let alone entire firms, produce a variety of products. Additionally, automobile and computer companies vertically integrate to themselves enter the battery industry. That being said, it is likely there exists considerable variation within each exporter that could influence how they react to MMC in foreign markets. For this reason, I also run specifications for firm and firm-year fixed-effects as a key robustness exercise to control for variation that stems from within firms themselves. In addition, as mentioned in Section 4, I also consider the same specifications for firms that only own a battery manufacturing establishment (approximately 150 firms) and all battery exporters (approximately 5,700 firms).

In addition, The LFTTD does not observe export activity form firms in third-party countries or domestic information on local producers. For instance, the LFTTD will capture all shipments from the U.S. exporting LIBs to Guatemala, but it does not have shipments from Japan. Table 9 provides descriptive statistics on these international battery markets. It is evident that U.S. exporters can face competition from exporters that are outside the scope of the data available. Firstly, 70% of battery imports into a market come from non-U.S. countries. Exporters located in foreign destinations are certainly a source of competition, but they themselves could have MMC with U.S. competitors leading to potential downward bias in measuring MMC. Therefore it is unclear what the effect of foreign party trade has on U.S. exporters' ability to collude in the market by exploiting MMC. It follows that the degree of bias in MMC_{imt} would naturally be higher if there is a larger share of trade coming

[32]https://www.lgchem.com/company/company-information/global-network/overseas-corporation/america shows LG Chem owns one manufacturing plant in Evansville, Indiana and three other subsidiaries around the United States.

from foreign countries in markets where firm i exports. Equation 4 shows the calculation for the *average share of foreign party trade* faced by firm i.

$$\overline{impShare}_{it}^{fpt} = \frac{1}{|M_{it}|} \sum_{m \in M_{it}} impShare_{mt}^{fpt} \mathbf{1}[Exports_{imt} > 0] \tag{4}$$

where $impShare_{mt}^{fpt}$ is the share of imports in market m that come from non-U.S. countries and M_{it} is the set of export markets for firm i at time t.[33]

Under the assumption that firm i will have more MMC with foreign exporters in market m if they export to markets with higher levels of foreign party trade, then $MMC_{imt}(1 + \overline{impShare}_{it}^{fpt})$ would better approximate the true level of MMC. Therefore including the interaction of MMC_{imt} with $\overline{impShare}_{it}^{fpt}$ would account for MMC with foreign exporters outside the U.S. Under my hypothesis that higher MMC leads to higher prices, I would expect the coefficients on the MMC_{imt} and the interaction to be positive and statistically significant.

Similar to competition from foreign parties, U.S. manufacturers could experience competition from non-manufacturing distributors (e.g. wholesalers). Roughly \$2.5 million of the approximately \$5.5 million of U.S. exports in batteries to an average market come from non-manufacturers. Although these firms tend to be much smaller and less active than firms in the sample (see Table 5), it's possible that a large presence of these types could hinder the ability of firms to leverage MMC across markets. To address this, I employ a similar strategy as I did above for foreign competition and calculate $\overline{NumNonManuf_{it}}$. I assume these distributors will be the noted earnings recipient when in fact they just passed on the price set by the firm; the true benefactor from exporting. Similar to the reasoning above, the presence of non-manufacturing exports is a source of market competition which would suggest a negative interaction, but could be positive if either the distributor or the manufacturer they work for have MMC with U.S. exporters. See Table 5 for a discussion of these

[33] I emphasize average share of foreign party trade because MMC is computed across markets where firm i is present. I also interact MMC with $impShare_{mt}^{fpt}$ to understand how foreign competition in the market itself impacts the effect of MMC on price.

firms.

As mentioned in section 3, LIB exporters have been shown to collude in non-battery product markets. For firms in the average market, I find that 61% of their revenue is earned in non-battery sectors (See Table 9). Firms that compete in both battery and non-battery markets will undoubtedly have MMC. To address this, I calculate the average value of non-battery exports for all U.S. battery exporters in market m ($\overline{NonBattExp_{mt}}$) and interact that total with MMC_{imt}. A positive interaction effect indicates that even firms that predominantly export non-battery products respond to MMC. On the other hand, a negative interaction effect and a positive effect on the stand-alone MMC coefficient suggests that battery dominant exporters are more sensitive to MMC. I expect interacting MMC with this proxy for the importance of battery sales to market competitors would mitigate the effect of MMC on export prices. That being said, the stand alone effect of MMC should be larger than that in the baseline result as the competitors that predominantly export batteries should be more sensitive to MMC.

5.4.4 Reverse Causation

While most of the MMC literature has found that firms with MMC tend to have higher earnings and behave more collusively, there is some evidence suggesting that firms deliberately seek out and use MMC to create suboptimal welfare equilibrium outcomes through mutual forbearance (Haveman and Nonnemaker (2000),Greve (2006),Baum and Korn (1999)). In addition, Scott (1991) finds that MMC is necessary for firms to coordinate in a concentrated industry which paves an argument for reverse causality in that firms disagreement in prices leads firms to seek MMC. As such, price setting behavior could actually explain entry decisions and MMC, as a result. This could leading indicate bias arising from reverse causality in my main specifications and should be addressed.

It should be emphasized that simply substituting a lagged measure for a concurrent one will not by itself sufficiently address the problem of reverse causality. The channel

of bias shifts from underlying relationships between independent variables to the (often unaddressed) question as to whether a dynamic relationship is appropriate (Bellemare et al. (2017), Reed (2015)). To address this possible endogeneity, I follow Reed (2015) and include a 2SLS specification using a lagged measure of MMC as an instrument. However, in doing this, one must be careful because even firms that are not exporting to the market can still have MMC with incumbent firms. As such, I make sure to calculate lagged MMC for firms even if they were not in the market in the previous period to ensure that the level of MMC in the previous period is fully captured. It follows from this argument that a lagged MMC value of 0 could indicate only one of two things. 1) The firm truly has MMC = 0 against incumbents in the market or 2) the firm did not export any batteries in the previous period or the market was not a U.S. export destination in the previous year. Therefore we must also control for whether there was a new entry in addition to the lagged MMC measure in the instrumenting equation. Equation 5.4.4 shows the IV regression I apply to the main specification described in Equation 1.

$$\begin{aligned} \text{First Stage: } Log(P_{imt}) &= \beta_{mmc}\mathbf{MMC}_{imt} + \beta_F\mathbf{F}_{it} + \beta_{FM}\mathbf{FM}_{imt} + \alpha_{im} + \gamma_t + \varepsilon_{imt} \\ \text{Second Stage: } \mathbf{MMC}_{imt} &= \mu + \gamma_1\mathbf{MMC_{im,t-1}} + \gamma_2 NE_{imt} + u \end{aligned} \tag{5}$$

The first stage is the main regression model from above (See Equation 1) and the second stage is the instrument equation. Here, μ is a constant, γ_1 is the effect MMC in period $t-1$ effects MMC in period t. NE_{imt} equals 1 if firm i was not an exporter at time $t-1$, but did export at time t or if market m was not an export destination at time $t-1$, but was a destination at time t for any firm. To assess the strength of the instrumental regression I first run a statistical test to check for endogenity in my main regressor MMC on log prices. If that test indicates endogeneity is present, I then run 2 additional tests to assess the strength of my instrumental equation defined in Equation 5.4.4. The first test is the Stock and Yogo (2005),

which provides information on whether the instrumental independent variables significantly explain the endogenous variable MMC. Also, noticing that I have more instruments than endogenous variables in my second stage regression, I also run a second test, called the Sargan-J test, to check for over-identification. The former test requires a significant result to ensure I am not using a weak instrument. For the latter test, I want to check that the result has lower P-value, which indicates that the instrumental variables are not correlated with the error term (and thus would need to be included in the second stage regression as well).

In addition, results from the first stage regression are also significant toward main motivating the main result. Significant results on lagged MMC in the first stage regression would suggest that MMC in a market for a firm is statistically stable and does not fluctuate significantly across time. This is important to establish since, fluctuation in the MMC parameter would suggest there is no long-term costs to competing more aggressively in marketsI expect this to be the case since exporters negotiate contracts or supply agreements that stipulate a unit price for a number of years. For example, a supply agreement between Panasonic (the manufacturer) and Tesla (the buyer) in July of 2009 suggests the amount and the unit price of LIBs were determined and set for a defined period of time. Hence, the assumption that suppliers predetermine the price they set in markets in future periods is not necessarily strong. I observe that the average firm sent between 19 to 20 shipments with a standard deviation of 182 shipments to the average market from 1997-2013, which suggests that firms in this sample repeatedly interacted within a given market.

5.4.5 Product Quality as an Alternate Mechanism

So far, I have asserted that the mechanism driving the effect between MMC and export prices is one of tacit collusion. An alternate theory could be one related to quality upgrading, where firms are better able to assess products sold by competitors with whom they have MMC and are better able to differentiate their product to better appeal to consumers. Using U.S.

Census import data Khandelwal (2010) finds heterogeneity in quality across the exporting country within each HTS-10 product in each time period. For this reason, I consider a regression specification with market-year fixed effects to account for the effect of quality on the export price of the product. Using a similar methodology, it's also possible that there could be heterogeneity in quality across firms exporting products in the same HS code. Therefore, I also consider a regression specification with firm-year effect effects.[34] I also run an ANOVA statistical test to show that the average unit values across markets are significantly different, which suggests that market boundaries as defined by country-HS6 pairs are clearly delineated.

6 Results

6.1 Main Result

My analysis shows that MMC has a positive and economically significant effect on U.S. export prices for different types of batteries. **Model 5** in Table 10 shows my main result that meeting a rival in one additional market increases export prices by 0.2 percent.[35] These results indicate that a one standard deviation increase in MMC leads to a 10 percent increase in U.S. export prices (see Table 6 for the summary statisitcs on MMC). From section 3, the European Commission found that consumers considered switching between different battery types in the face of a 5 to 10% increase in price indicating that these effects are economically significant. Furthermore, this coefficient magnitude is similar to past studies, most notably Ciliberto and Williams (2014) when they assume market and year fixed effects. Coefficient standard errors hardly change when using the feasible GLS estimators with market-year and firm-year random effects in **Model 1** and **Model 3**, respectively. nor does the sign change

[34]To fully account for quality, one could first compute quality-adjusted prices before analyzing effects of MMC. Insignificant coefficient estimates would provide evidence for this theory, where robust estimates would provide evidence for the tacit collusion framework. I do not address this approach in this paper, but it is a worthwhile pursuit for future research.

[35]Table 10 shows the effect of increasing MMC by 10.

when considering market-year fixed effects (**Model 1** and **Model 2**). Coefficient magnitudes increase when one assumes firm-year fixed effects as shown in **Model 4**, suggesting that omitted firm-year level variables downward bias the coefficient estimates. The general result does not change in **Model 5** when replacing concurrent MMC measures with a lagged measure. The χ^2 from a Hausman test between **Model 5** and **Model 6** was approximately 0, which suggests a rejection of the null hypothesis that a random effects model is appropriate at a less than 1% level of significance. Furthermore, using Firm-Market and Year effects best had higher overall R^2 than other specifications. The models considered in Table 11 suggests a robust result despite the suite of different control variables I considered along with fixed effects at different levels.

6.2 Robustness Checks

6.2.1 Omitted Variable Bias

The effect of MMC is robust to several different specifications and assumptions. **Models 1 - 4** in Tables 11 and 10 shows a consistently positive and significant effect of MMC across specifications with differing sets of fixed and random effects. Estimates suggest the effect from a 1 unit increase in MMC ranges from .2 percent if we include year, firm, product and country fixed-effects to roughly .8 percent if we assume all random-effects. **Model 2** shows that specification with market-year fixed effects shows an effect of .019, which is approximately equal to the coefficient in my preferred model. Including market-year fixed-effects helps control for any omitted variable bias to market concentration (see discussion in 5.4). With similar coefficient magnitudes across both models, excluding market concentration from the regression does not bias the effect of MMC on price in a significant way in this setting.

6.2.2 Different Sets of Competitors

Tables 12 and 13 show similar analyses to Tables 11 and 10, respectively, but with set of all battery exporters. Interestingly, if you make the assumption that all battery exporters

strategically interact in overseas markets, the coefficients on MMC are still significant and positive and comparable when I use the set of exporters with a manufacturing establishment.

However, if one restricts the sample of competitors to battery exporters with a *battery* manufacturing plant like in Tables 14 and 15, then the coefficients remain positive and significant, but are orders of magnitude bigger than their comparable coefficients in the baseline analysis. For instance, in the preferred specification with market-year fixed-effects (**Model 2** in Table 15) an increase in MMC by 10 increases price by almost 4%, which is double the effect seen in **Model 5** in Table 10.[36]

Although it is possible that battery manufacturers can compete against firms that do not primarily manufacture batteries, it is more likely that battery manufacturers compete among themselves. Hence, the increase in coefficient magnitude after restricting the sample to the roughly 150 battery manufacturers operating in the United States is further support of my initial hypothesis that MMC leads to more collusive outcomes in the global battery market.

6.2.3 Exports-At-Risk

Up until now, I have implicitly assumed that firms place equal weight to each contact it has with competitors in other markets. However, it is also plausible that firms weight their contacts differently depending on how important other markets besides market m are to firm i. The more contact a firm has with competitors in other markets, the more its sales in other markets are at risk if it decides to behave aggressively by slashing prices (Feinberg (2015)).[37] Referencing Table 16, **Model 1** shows a 1% increase in XAR for firm i in market m corresponds with a 1.5% increase in firm i's export price to market m. Coefficient signs are similar to those found for specifications using non-weighted measures of MMC (Table 11). These results are consistent in magnitude and direction to those reported in Feinberg (2015).

[36] Unfortunately with the small sample size, future analysis using this sample will be difficult.
[37] See the appendix in Chapter 1 for details on the computation of XAR.

6.2.4 External Competition

As mentioned in Section 5.4, interacting foreign competition on MMC sheds light on the ability of U.S. firms to maintain above equilibrium prices when faced with competition from non-U.S. exporters and also corrects for potential measurement in MMC. **Model 1** in Table 18 interacts the share of foreign party trade with MMC and **Model 2** the average share of foreign party faced in all of firm i's export markets ($\overline{impShare}_{it}$) with MMC. To compute MMC for firm i in market m, one needs information on market structure for all markets where firm i is present. As $\overline{impShare}_{it}$ approaches unity, the coefficient of MMC increases from .005 to .01. As stated in Section 5.4, one reason the coefficient effect is positive could be that foreign competitors themselves have MMC with firms in the sample (i.e. downward bias in MMC measurement.)

Furthermore, in **Model 2** the coefficient on MMC slightly decreases from roughly .1 when the firm faces mostly foreign competition in it's export markets compared to .05 indicating that MMC has a weaker effect on price when foreign rivals are not present. This is an interesting result that deserves further exploration, but one theory could be that U.S. firms band together with well-known competitors to combat foreign competition. However, a perhaps more likely theory is that major exporters in the battery industry are exporting from foreign destinations and also have MMC with U.S. exporters.

I replicate the above approach in **Model 3** and **Model 4** except I substitute the count of out-of-sample non-manufacturing battery exporters for share of trade from foreign parties. The interaction effects were either insignificant (**Model 3**) or marginal (**Model 4**) suggesting that unobserved MMC due to the presence of non-manufacturing exporters matters little (and is perhaps independent from observed MMC) to estimating the coefficient on MMC.

In **Model 5** and **Model 6**, I include the interaction of the share of firm i's non-battery exports (**Model 5**) and the average share of non-battery exports for rivals in market m (**Model 6**) with MMC. MMC has a larger effect for firms that mostly export batteries and/or markets whose rivals mostly export batteries. Looking at **Model 6**, when the average share

of total exports in non-batteries is close to 0 for all firms in the market, then meeting a competitor in one more market is linked to more than a 1.5% increase in price. That being said, in **Model 6**, even in markets where incumbents mostly export non-battery products, the net effect of MMC on price still remains positive ($0.177 - 0.139 = .038$). Both coefficients are negative and significant indicating that the downward bias introduced from potential MMC across battery and non-battery markets is a source of inflated coefficient estimates for MMC.

In **Model 7**, I combine all the interaction terms into one model to more completely control for the extent of downward bias in the model. Generally, in markets where firms do not face much foreign competition and also mostly export products other than batteries, MMC could negatively effect prices.[38] However, MMC has the anticipated positive effect on price for more prominent battery exporters and the effect is stronger if the firm regularly faces foreign competition.

6.2.5 Reverse Causation

Finally, it is possible that there is some reverse causation where export prices themselves determine market entry decisions and MMC, as a result (see Section 5.4 for more discussion). Table 17 shows the results from running a two-stage least squared regression with Equation 5.4.4 as the main instrumental regression. The results show that endogeneity between MMC and prices do exist, and that the instrumental regression is valid. The P-value from Stock and Yogo test show that the instrumental regression significantly explains MMC and the insignificant Sargan-J statistic suggests variables used in that equation are not correlated with the error term. As such, I can assume the first stage regression is valid as an instrument and can use it to predict MMC for second stage regression. The results from that second stage regression suggest that an increase in MMC by 10 leads to a 4 percent increase in price, which is in line with the coefficients estimated from other models discussed. Furthermore,

[38] It should be noted that such cases should be more thoroughly examined by including potential omitted variable bias not used in this analysis.

it suggests that the bias from reverse causality in the main specification is likely downward, which means the true effect is likely larger than the 2 percent highlighted above.

7 Conclusion

I find that battery exporters tend to charge higher prices when faced with MMC using a combination of confidential international trade transaction and establishment level data provided by the U.S. Census Bureau. The main result also does not change when assuming different fixed-effects specifications, implemented different measures of MMC commonly considered in the literature while controlling for a myriad of firm, market and firm-market level control variables. I find a one standard deviation increase in the degree of MMC significantly corresponds to over a 1% increase in market price. This result is consistent with other empirical results spanning across the literature.

Motivated by evidence of collusion between battery manufacturers in the U.S. and elsewhere, the broad objective of this paper is to use simple econometric methodology to show the existence of a robust relationship between MMC and export prices in an international trade setting. From here, there are many paths of future research that can be taken to show how MMC impacts firm export decisions. Besides studying this question using a different industry, it would be beneficial to implement a more structural model that captures the dynamic interplay between price and MMC that is more in line with the theoretical literature. Parallel to this point, one could use the granular nature of shipment level data to observe whether and exactly how firms retaliate when faced with an aggressive competitor. However, leading into that, one must first think about how firm's entry into global markets is impacted by MMC. Finally, the majority of existing theory assumes two-firm, two-market models but in reality there are many firms competing in many, but not all markets. Such rich market structure could have network type effects on firm behavior. Finally, more exploration of the mechanisms driving the competitive effects of MMC could convey that certain types of

MMC can actually be welfare enhancing as opposed to universally detrimental.

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8 Declarations

8.1 Author Information

I am an independent researcher with interests in international trade and applied game theory. I currently work as an economist for a U.S. federal agency. I received my PhD in economics from American University with a focus in international trade and industrial organization. I wrote this paper in my personal capacity. All thoughts expressed in this paper are my own and do not reflect those of the U.S. government. I was a part-time Census Bureau employee and a graduate student at American University when I started this research.

8.2 Funding Information

This research received no specific grant or funding from any funding agency in the public, commercial, or not-for-profit sectors.

8.3 Ethics Declaration

*Any opinions and conclusions expressed herein are those of the author and do not represent the views of the US Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7532452: CBDRB-FY24-0445). I thank my dissertation committee members: Robert Feinberg, Nathan Larson and Alan Isaac. In addition, I would like to thank Fariha Kamal, Emek Basker, Emin Dinlersoz and Mary Hansen for valuable advice and counsel; Shawn Klimek for his guidance and direction in finding materials needed to complete this research. I also thank Michael Freiman for timely disclosure avoidance assistance. All mistakes and omissions are my own. I declare that I have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

9 Tables and Figures

Example of Multi-Market Competition

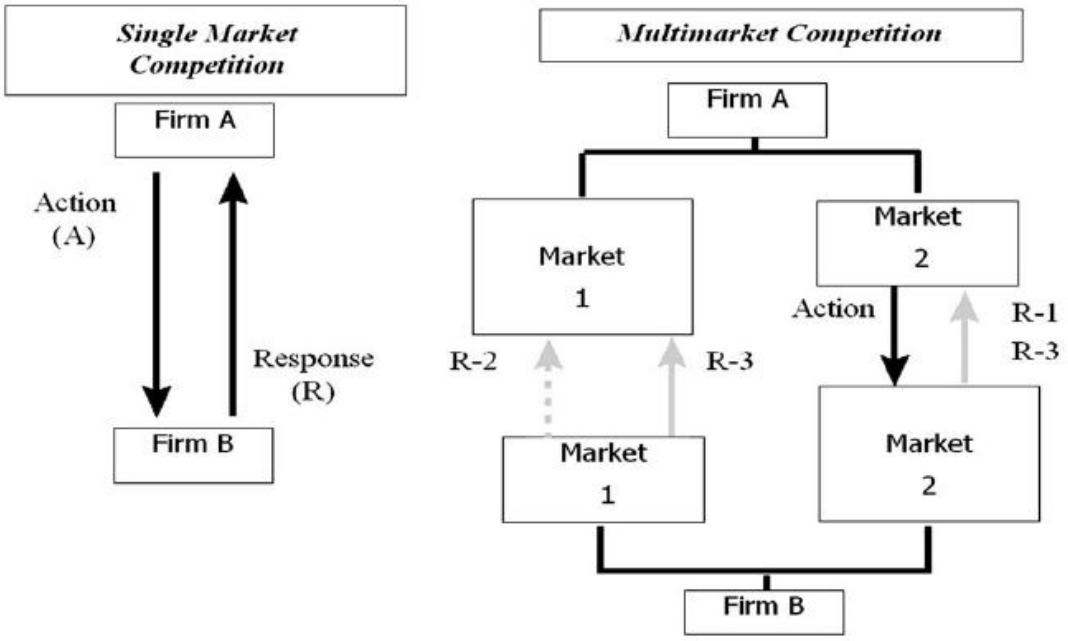


Figure 1: On the left: Two firms, A and B competing in one market. On the right: A and B are competing in market 1 and market 2. Firm A has sphere of influence in Market 1 and Firm B has sphere of influence in Market 2. If Firm A wants to attack firm B in market 2 then Firm B can retaliate in market 1 (a market firm A values), in market 2 or both at the same time. This fear of cross-market retaliation makes both firms more hesitant to engage each other competitively. Figure is courtesy of Stephan and Peters (2013)

Prisoner's Dilemma Game Among Competitors

		Firm 2	
		C	D
Firm 1	C	$\frac{\Pi^c}{2}$, $\frac{\Pi^c}{2}$	0, Π^d
	D	Π^d, 0	Π_1^N, Π_2^N

Table 1: Example of Prisoner's Dilemma game. Nash Equilibrium is where both firms choose D. The collusive outcome occurs when both firms choose C.

Prisoner's Dilemma Game Assuming "Unlinked" Strategies

Market A

		Firm 2: C	Firm 2: D
Firm 1	C	$\frac{\Pi_A^c}{2}, \frac{\Pi_A^c}{2}$	$0, \Pi_A^d$
	D	$\Pi_A^d, 0$	$\Pi_{1,A}^N, \Pi_{2,A}^N$

Market B

		Firm 2: C	Firm 2: D
Firm 1	C	$\frac{\Pi_B^c}{2}, \frac{\Pi_B^c}{2}$	$0, \Pi_b^d$
	D	$\Pi_B^d, 0$	$\Pi_{1,B}^N, \Pi_{2,B}^N$

Table 2: Firms competing across two markets will play two, possibly asymmetric, Prisoner's Dilemma games.

Prisoner's Dilemma Game Assuming "Linked" Strategies

		Firm 2: C	Firm 2: D
Firm 1	C	$\frac{\Pi_A^c}{2} + \frac{\Pi_B^c}{2}, \frac{\Pi_A^c}{2} + \frac{\Pi_B^c}{2}$	$0, \Pi_A^d + \Pi_B^d$
	D	$\Pi_A^d + \Pi_B^d, 0$	$\Pi_{1A}^N + \Pi_{1B}^N, \Pi_{2A}^N + \Pi_{2B}^N$

Table 3: Players pool incentive constraints across markets and play one single game. Their actions then apply to both markets.

U.S. Exports in Batteries from 1992 - 2016

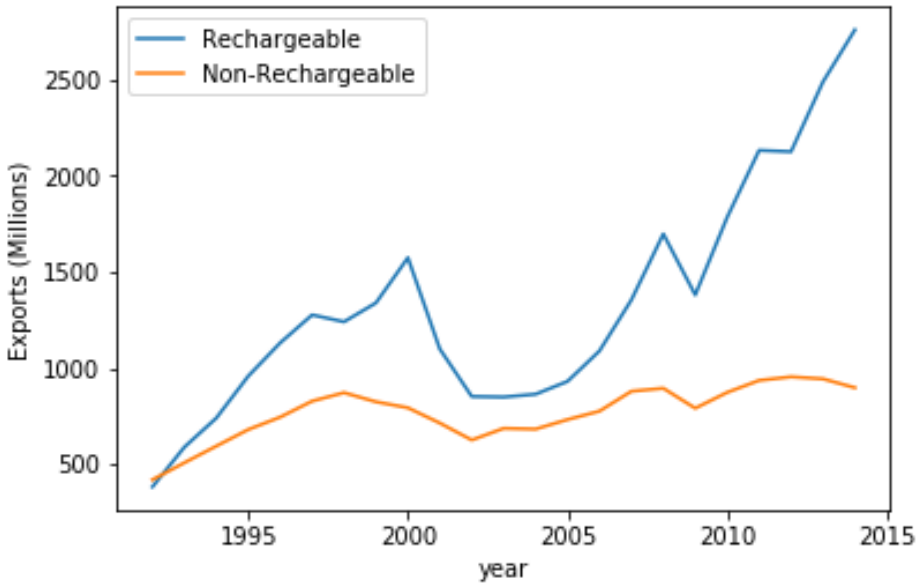


Figure 2: Source: UN Comtrade Database.

Number of U.S. Battery Importing Countries from 1992 - 2016

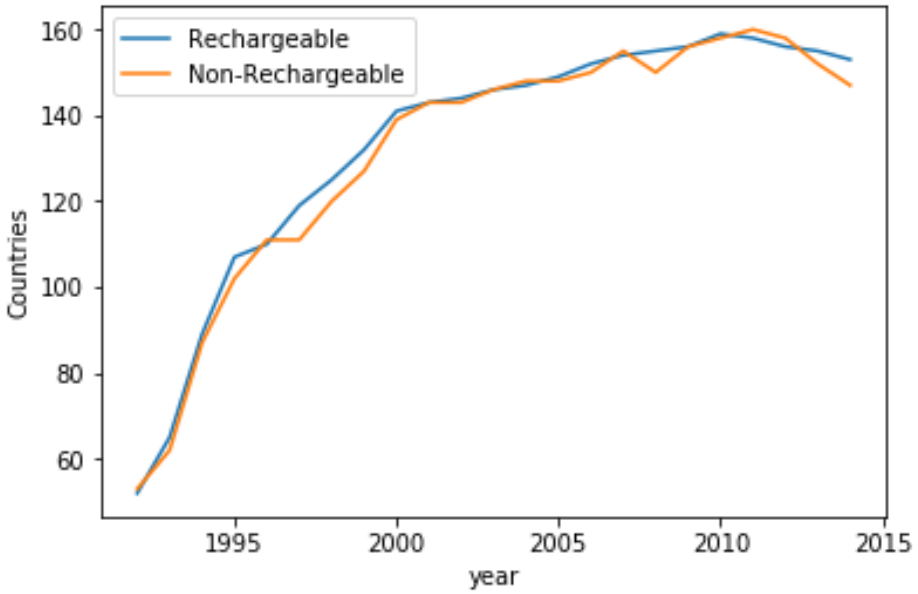


Figure 3: Source: UN Comtrade Database.

Product Segmentation in Battery Industry

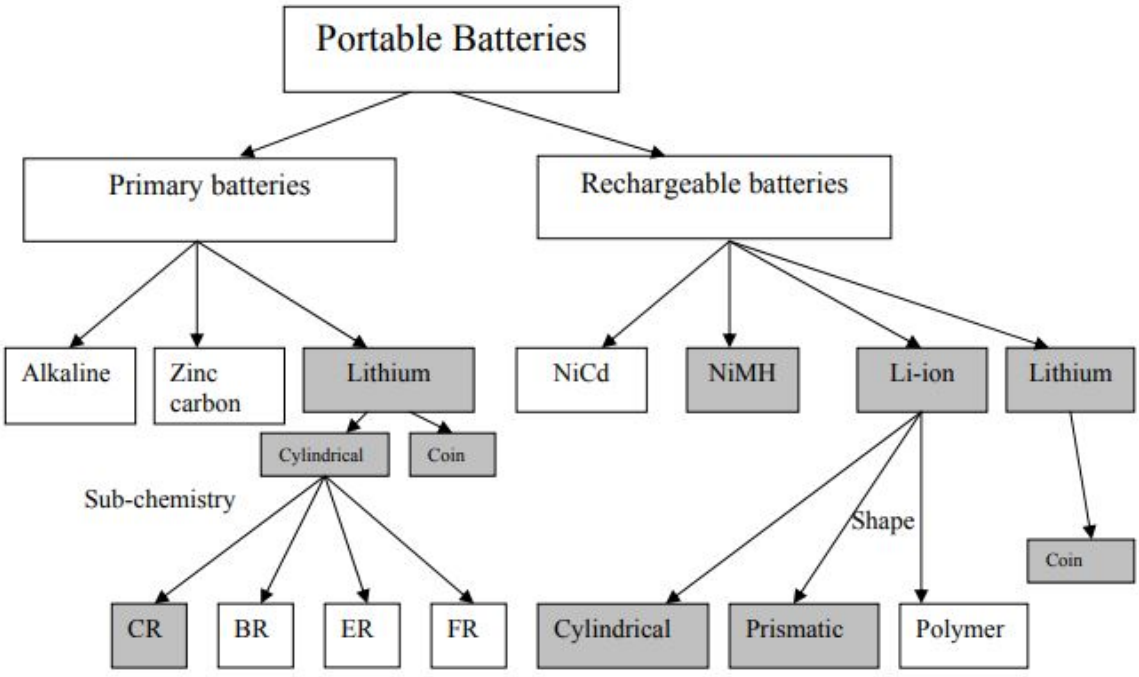


Figure 4: Battery Market Segmentation suggested by the EC. The EC finds that market segmentation occurs at the sub-chemistry level. Shaded areas show the overlap between Panasonic and Sanyo. CR means Lithium manganese dioxide, BR means Lithium poly-carbon monofluoride, ER means Lithium thionyl chloride and FR means Lithium-Iron Disulfide (Council of European Union (2009)).

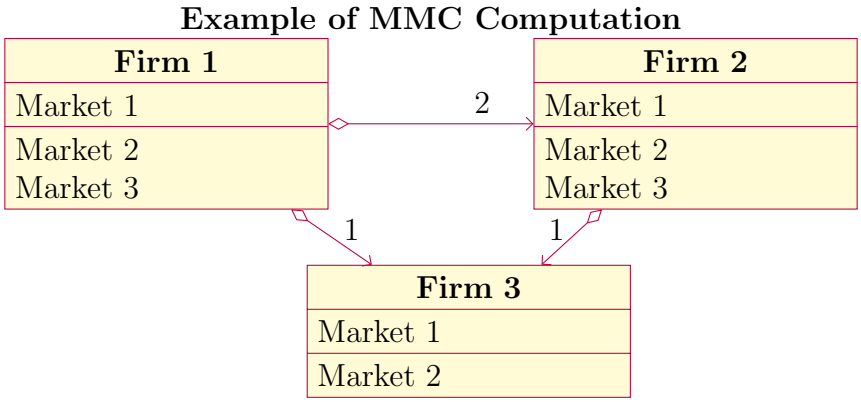


Figure 5: Firm 1 faces firm 2 in 2 other markets besides market 1 and firm 3 in one other market. Hence, MMC for firm 1 in market 1 is 2 + 1 = 3.

Portrait of Firms in the Sample

Variable	Avg.	Std. Dev.	Firm-Years
Total Firm Battery Exports (USD) (i,t)	3,582,000	16,550,000	5,500
Number of Manuf. Estabs. (i,t)	12.7	27.2	5,500
Firm Employment (i,t)	12,160	46,530	5,500
Number of Other Export Markets (i,t)	5.39	13.77	5,500
% Total Exp. in Batt (i,t)	.21	0.35	5,500
Total Non-Batt. Exports (i,t)	4,176,000	19,800,000	5,500
Avg. % Imports from FPT (i,t)	.66	.24	5,500

Table 4: These are firm-level control variables. "Number of other exports markets" counts the total number of alternative export markets for a firm. For instance, if the a firm has 3 export markets, then it will have 2 other markets relative to any one export market. "% Total Exp. in Batt" is firm i's total share of export revenue earned from battery sales. "Total Non-Batt. Exports" is firm i's export revenue earned in the sale of electronics that use batteries (e.g. laptops, automobiles). "Avg. % Imports from FPT" is the average share of imports from non-U.S. countries in firm i's export markets. In the average year there are $\frac{5500}{16} = 350$ battery exporters that own a manufacturing establishment. A firm is a battery exporter if it sends at least one battery shipment to a foreign market.

Non-Manufacturing Battery Exporters

Variable	Avg.	Std. Dev.	Firm-Years
Battery Export sales	800,100	8,104,000	30,000
Firm Employment	4,394	29,780	30,000

Table 5: Description of battery exporters that do not have a manufacturing establishment. In the average year there are $\frac{30000}{16} = 1875$ battery exporters that do not own a manufacturing establishment.

Descriptives on MMC

Market Definition	Stat	MMC	Mkt. Competitors	Obs.
Ctry-HS4	Avg.	103.80	6.42	25,000
	Std. Dev.	125.50	10.19	25,000
Ctry-HS6	Avg.	35.23	2.51	29,500
	Std. Dev.	49.10	4.67	29,500

Table 6: MMC is a pooled average from panel data unique at the year-firm-market level. In this table and all others below, observations are rounded and all values have at most 4 significant digits in accordance with U.S. Census Bureau disclosure policy.

Manufacturing Battery Exporters from 1997 - 2013

Variable	Avg.	Std. Dev.	Med.
Intra-Firm Exports (USD) (i,m,t)	204,300	2,119,000	1,139
Arm-Length Exports (USD) (i,m,t)	357,500	2,585,000	44,300
Total Exports (USD) (i,m,t)	561,800	3,727,000	54,890
Quantity (units) (i,m,t)	168,700	2,106,000	847.70
Price (AL Revenue/Qty.) (i,m,t)	**906.40**	**8,405.00**	**53.33**

Table 7: Trade activity for firm (i) in the average export market (m) in the average year (t). Medians are interpolated by averaging some values above and below the true median in accordance with Census Bureau Disclosure Policy.

Extent of Access to Foreign Markets

Variable	Avg.	Std. Dev.
Shr. of intra-firm export value	0.11	0.26
Experience In Market (years)	3.78	3.36

Table 8: "Pct. of intra-firm export value" is the share of intra-firm shipments by firm i to market m. "Experience In Market" is the number of past years firm i was exporting to market m.

Characteristics of Export Markets

Variable	Avg.	Std. Dev.	Market-Years
Total U.S. Exports to a Market (m,t)	3,385,000	15,630,000	8,000
Total Exports by Non-Manufacturers (m, t)	2,421,000	13,150,000	8,000
Number of Non-manuf exporters in market (m,t)	9.21	18.02	8,000
gdp per capita	21,780.00	2,071.00	8,000
Imports from FPT (m,t)	33,090,000	156,700,000	8,000
Share of imports from FPT (m,t)	0.70	.35	8,000
Product Exports from Dest. Country (m,t)	35,160,000	205,700,000	8,000
Mkt. Avg. % of Non-Batt Exp. (m,t)	.61	.33	8,000

Table 9: Markets are defined as country-HS6 pairs. "Number of Non-manuf exporters in market" is the number of non-manufacturers that export to market m and "Total Exports by Non-Manufacturers" is the value they export to market m. " Share of imports from FPT" is the share of imports from firms located in non-U.S. countries and "Imports from FPT" is the amount they export. "Product Exports from Dest. Country" is the value of exports from potential competitors in the foreign market itself. "Mkt. Avg. % of Non-Batt Exp." is the share of revenue earned in non-battery markets among U.S. firms in export market m.

Robust Relationship between MMC and Log Battery Export Prices

Covariates	Model 1 (GLS)	Model 2	Model 3 (GLS)	Model 4	**Model 5 (Fixed Effects)**	Model 6 (Random Effects)
MMC * 10^{-1} (i,m,t)	0.031***	0.019***	0.035***	0.039***	**.020*** **	.013**
	[0.003]	[0.003]	[0.004]	[0.004]	**[.004]**	[.004]
MMC * 10^{-1} (i,m,t-1)					**0.13*** **	0.011**
					[0.003]	[0.004]
R-Sq. Overall	0.212	0.192	0.164	0.156	**0.207**	0.105
Obs.	29,500	29,500	29,500	29,500	**29,500**	29,500
Effects	Market-Year RE	Market-Year FE	Firm-Year FE	Firm-Year FE	**Firm-Market, Year FE**	Firm-Market, Year RE
Clustering By	MARKET-YEAR	MARKET-YEAR	FIRM-YEAR	FIRM-YEAR	**Firm-Market**	Firm-Market

Table 10: Model 5 shows my main result. The control variables shown in Table 11 were included in implementation, but are suppressed from the output. The χ^{sq} from a Hausman tests was approximately 0, which suggests a rejection the null hypothesis that a random effects model is appropriate at a 1% level. Coefficient standard errors are in brackets. $***p < .001$, $**p < .01$, $*p < .05$, $+p < .10$

The Effect of MMC on Log Battery Export Prices

Covariates	Model 1	Model 2	Model 3	Model 4
$MMC * 10^{-1}(i,m,t)$	*0.078****	*0.029****	*0.039****	*0.018****
	[0.003]	*[0.003]*	*[0.003]*	*[0.003]*
Shr. of Shipments to RP (i,m,t)	-2.949***	-2.576***	-2.410***	-2.349***
	[0.045]	[0.040]	[0.043]	[0.040]
Experience In Market	-0.028***	-0.016***	-0.034***	-0.031***
	[0.003]	[0.003]	[0.003]	[0.003]
Log GDP per capita (m,t)	-0.097***	-0.05	-0.087***	-0.084
	[0.011]	[0.110]	[0.010]	[0.095]
Log Imports from Third-Party Countries (m,t)	-0.072***	-0.013	-0.048***	-0.006
	[0.006]	[0.008]	[0.006]	[0.006]
Log Product Exports from Dest. Country (m,t)	0.015***	-0.005	-0.005	-0.001
	[0.003]	[0.006]	[0.003]	[0.004]
Number of Other U.S. Exporters (m,t)	0.014***	-0.001	0.013***	-0.008**
	[0.003]	[0.003]	[0.003]	[0.003]
Number of Non-manuf exporters in market (m,t)	-0.004***	0.005***	-0.004***	0.003**
	[0.001]	[0.001]	[0.001]	[0.001]
Log Avg. Non-battery Exp. (m,t)	0.096***	0.022**	0.027***	0.007
	[0.007]	[0.007]	[0.006]	[0.006]
Avg. % Exp in non-Batteries (m,t)	-0.225**	-0.127+	-0.043	-0.148*
	[0.071]	[0.077]	[0.064]	[0.065]
Log Total Exports (m,t)	-0.038**	0.031+	0.011	0.019
	[0.012]	[0.017]	[0.010]	[0.012]
Share of imports from FPT (m,t)	0.858***	-0.119	0.782***	-0.021
	[0.085]	[0.126]	[0.074]	[0.080]
Number of Other Markets Serviced (i,t)	-0.002***	0	-0.001	-0.002**
	[0.001]	[0.000]	[0.001]	[0.001]
Log Firm employment (i,t)	0.065***	0.068***	0.109**	0.072*
	[0.008]	[0.007]	[0.034]	[0.031]
Log Non-Battery Exports (i,t)	-0.029***	-0.012*	-0.017+	-0.012
	[0.006]	[0.005]	[0.009]	[0.008]
% Export Value in Batteries (i,t)	-0.486***	-0.309***	0.059	-0.01
	[0.071]	[0.062]	[0.141]	[0.130]
Log Firm Exports (i,t)	-0.143***	-0.130***	-0.105***	-0.098***
	[0.009]	[0.008]	[0.017]	[0.015]
Avg. FPT Imp Share faced by firm (i,t)	0.864***	0.465***	0.230*	0.316**
	[0.085]	[0.076]	[0.109]	[0.101]
R-Sq. Overall	0.223	0.174	0.2	0.401
Obs.	29500	29500	29500	29500
Year Effects	FIXED	FIXED	FIXED	FIXED
Firm Effects	NONE	NONE	FIXED	FIXED
Country Effects	NONE	NONE	NONE	FIXED
Product Effects	NONE	NONE	NONE	FIXED
Market Effects	NONE	FIXED	NONE	NONE
Clustering By	Market-Year	Market-Year	Market-Year	Market-Year

Table 11: (i,m,t) refers to a covariate that varies by the firm, market and year, (m,t) refers to a covariate that varies by market and year and (i,t) refers to a covariate that varies by firm and year. Coefficient standard errors are in brackets. $***p < .001$, $**p < .01$, $*p < .05$, $+p < .10$

Baseline Results: Sample = All U.S. battery exporters

Covariates	Model 1	Model 2	Model 3	Model 4
$MMC * 10^{-1}(i,m,t)$	0.062***	0.020***	0.031***	0.016***
	[0.003]	[0.002]	[0.002]	[0.002]
R-Sq. Overall	0.174	0.119	0.15	0.345
Year Effects	FIXED	FIXED	FIXED	FIXED
Firm Effects	NONE	NONE	FIXED	FIXED
Country Effects	NONE	NONE	NONE	FIXED
Product Effects	NONE	NONE	NONE	FIXED
Market Effects	NONE	FIXED	NONE	NONE
Market * Year Effects	NONE	NONE	NONE	NONE
Firm * Year Effects	NONE	NONE	NONE	NONE
Clustering By	NONE	NONE	NONE	NONE
Firms	5,700	5,700	5,700	5,700
Markets	1,200	1,200	1,200	1,200
Firm-Years	14,000	14,000	14,000	14,000
Market-years	8,900	8,900	8,900	8,900
Obs	45,500	45,500	45,500	45,500

Table 12: Replicating the baseline results in Table 11, but with the set of all battery exporters.

Baseline Results Continued: Sample = All U.S. battery exporters

Covariates	Model 1: RE	Model 1: FE	Model 2: RE	Model 2: FE	Model 3: FE
$MMC * 10^{-1}(i,m,t)$	0.018***	0.009***	0.028***	0.030***	0
	[0.002]	[0.002]	[0.003]	[0.004]	[0.003]
R-Sq. Overall	0.162	0.139	0.138	0.129	
Year Effects	NONE	NONE	NONE	NONE	NONE
Firm Effects	NONE	NONE	NONE	NONE	NONE
Country Effects	NONE	NONE	NONE	NONE	NONE
Product Effects	NONE	NONE	NONE	NONE	NONE
Market Effects	NONE	NONE	NONE	NONE	NONE
Market * Year Effects	RANDOM	FIXED	NONE	NONE	FIXED
Firm * Year Effects	NONE	NONE	RANDOM	FIXED	FIXED
Clustering By	MARKET-YEAR	MARKET-YEAR	FIRM-YEAR	FIRM-YEAR	Mkt-YEAR
Firms	5,700	5,700	5,700	5,700	5,700
Markets	1,200	1,200	1,200	1,200	1,200
Firm-Years	14,000	14,000	14,000	14,000	14,000
Market-years	8,900	8,900	8,900	8,900	8,900
Obs	45,500	45,500	45,500	45,500	45,500

Table 13: Same analysis as Table 10, but with all battery exporters.

Baseline Results: Sample = Battery Manufacturers

Covariates	Model 1	Model 2	Model 3	Model 4
$MMC * 10^{-1}(i,m,t)$	0.124**	0.029***	0.028***	0.005
	[0.008]	[0.008]	[0.007]	[0.007]
R-Sq. Overall	0.268	0.23	0.212	0.48
Year Effects	FIXED	FIXED	FIXED	FIXED
Firm Effects	NONE	NONE	FIXED	FIXED
Country Effects	NONE	NONE	NONE	FIXED
Product Effects	NONE	NONE	NONE	FIXED
Market Effects	NONE	FIXED	NONE	NONE
Market * Year Effects	NONE	NONE	NONE	NONE
Firm * Year Effects	NONE	NONE	NONE	NONE
Clustering By	NONE	NONE	NONE	NONE
Firms	150	150	150	150
Markets	850	850	850	850
Firm-Years	700	700	700	700
Market-years	5,400	5,400	5,400	5,400
Obs	11,000	11,000	11,000	11,000

Table 14: Replicating the baseline results in Table 11, but with the set of all battery manufacturers.

Baseline Results Continued: Sample = Battery Manufacturers

Covariates	Model 1: RE	Model 1: FE	Model 2: RE	Model 2: FE	Model 3: FE
$MMC * 10^{-1}(i,m,t)$	0.077***	0.037***	0.023**	0.026**	-0.001
	[0.007]	[0.009]	[0.008]	[0.008]	[0.009]
R-Sq. Overall	0.257	0.212	0.225	0.218	
Year Effects	NONE	NONE	NONE	NONE	NONE
Firm Effects	NONE	NONE	NONE	NONE	NONE
Country Effects	NONE	NONE	NONE	NONE	NONE
Product Effects	NONE	NONE	NONE	NONE	NONE
Market Effects	NONE	NONE	NONE	NONE	NONE
Market * Year Effects	RANDOM	FIXED	NONE	NONE	FIXED
Firm * Year Effects	NONE	NONE	RANDOM	FIXED	FIXED
Clustering By	MARKET-YEAR	MARKET-YEAR	FIRM-YEAR	FIRM-YEAR	Mkt-YEAR
Firms	150	150	150	150	150
Markets	850	850	850	850	850
Firm-Years	700	700	700	700	700
Market-years	5,400	5,400	5,400	5,400	5,400
Obs	11,000	11,000	11,000	11,000	11,000

Table 15: Same analysis as Table 10, but with all battery manufacturers.

Exports-At-Risk

Covariates	Model 1
$Log(XAR)$ (i,m,t)	0.015***
	[0.003]
Fixed-Effects	Market-Year
Clustering By	MARKET-YEAR

Table 16: Model 1 has XAR is a weighted alternative to MMC used in Table 11. For reference, average XAR is $\$47,450,000$ with a standard deviation of $\$152,800,000$. Coefficients from controls reference in Table 11 were used to run this model and suppressed. Coefficient standard errors are in brackets. $***p < .001$, $**p < .01$, $*p < .05$, $+p < .10$

Addressing Reverse Causation: Two Stage Least Squares

Covariates	Second Stage (Dependent Variable: $log(P_{imt})$	First Stage (Dependent Variable: MMC_{imt})
MMC (i,m,t)	0.040***	
	[0.008]	
MMC (i,m,t-1)		.979***
New Entrant (i,m,t)		1.129***
Test for Endogeneity F Statistic	67.5	
Test for Endogeneity P-Value	$< .01$	
Stock and Yogo P-Value	$< .01$	
Sargan-J Statistic P-Value	0.90	
Fixed Effects	Firm-Market and Year	None
Clustering By	MARKET-YEAR	None

Table 17: The results in the first stage regression (right column) show the effect of MMC in the previous period and whether the firm is a new exporter to the market is a new destination not observed in the data in the previous period. The second stage regression (left column) shows the effect of MMC on log prices using the first stage regression as an instrument. To assess the validity of the instrument, I include tests for endogeneity, a test for whether the instrument is strong (significantly explains the the endogenous variable) and whether the instrumental variables in the first stage regression are correlated with the error term. Statistical significance for the endogeneity tests and the Stock and Yogo test suggests a strong instrument, while and a statistically insignificant result for the Sargan-J Statistic to shows that these variables are not a source omitted variable bias in the second stage regression. Coefficients for control variables displayed in Table 11 are suppressed in displaying the output, but used in calculating the main coefficient. $***p < .001$, $**p < .01$, $*p < .05$, $+p < .10$

Effects of External Competition

Covariates	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7
MMC * 10^{-1} (i,m,t)	0.066***	0.050**	0.102***	0.081***	0.090***	0.177***	0.103***
	[0.007]	[0.017]	[0.005]	[0.005]	[0.005]	[0.012]	[0.022]
Number of Non-manuf. exporters in market (m,t)	-0.004*	-0.005**	-0.003	-0.005**	-0.005**	-0.005**	-0.003+
	[0.002]	[0.002]	[0.002]	[0.002]	[0.002]	[0.002]	[0.002]
Average % of Non-Battery Exports by rivals (m,t)	-0.206*	-0.189*	-0.210*	-0.193*	-0.220*	0.096	0.123
	[0.094]	[0.094]	[0.094]	[0.094]	[0.094]	[0.097]	[0.101]
Share of imports from FPT (m,t)	0.661***	0.787***	0.758***	0.786***	0.792***	0.848***	0.744***
	[0.138]	[0.129]	[0.129]	[0.129]	[0.129]	[0.128]	[0.138]
Avg. FPT Imp Share faced by firm (i,t)	1.164***	1.029***	1.106***	1.099***	1.112***	1.089***	1.001***
	[0.112]	[0.117]	[0.109]	[0.109]	[0.109]	[0.109]	[0.124]
Avg. Non-Manuf Exp. (i,t)	0.004***	0.004***	0.003***	0.004***	0.004***	0.004***	0.003***
	[0.001]	[0.001]	[0.001]	[0.001]	[0.001]	[0.001]	[0.001]
% Export Value in Non-Batteries (i,t)	0.464***	0.460***	0.472***	0.469***	0.529***	0.501***	0.487***
	[0.069]	[0.069]	[0.069]	[0.069]	[0.077]	[0.069]	[0.077]
MMC * FPT IMP. SHR * 10^{-1}	0.029**						0.022+
	[0.009]						[0.011]
MMC * Avg. FPT IMP. SHR * 10^{-1}		0.048*					0.087***
		[0.023]					[0.024]
MMC * Num. Non-Manuf Exporters * 10^{-1}			-0.000***				-0.000*
			[0.000]				[0.000]
MMC * Avg. Non-Manuf Exp. * 10^{-1}				0			0.000**
				[0.000]			[0.000]
MMC * % NonBatt Exp. * 10^{-1}					-0.010+		-0.002
					[0.005]		[0.006]
MMC * Avg. % Non-Batt Exp * 10^{-1}						-0.139***	-0.153***
						[0.016]	[0.018]
R-Sq. Overall	0.23	0.23	0.23	0.23	0.23	0.23	0.24
Obs.	29500	29500	29500	29500	29500	29500	29500
Year Effects	FIXED	FIXED	FIXED	FIXED	FIXED	FIXED	FIXED
Clustering By	MARKET-YEAR	MARKET-YEAR	MARKET-YEAR	MARKET-YEAR	MARKET-YEAR	MARKET-YEAR	MARKET-YEAR

Table 18: Models 1 through 7 control for the extent of external competition on the ability of firms to use MMC to maintain high export prices. Controls in Table 11 are not included in this analysis in order to more fully understand the relationships seen in the data. Models 1 and 2 are concerned with competitors from foreign countries. Models 3 and 4 test the extent of downward bias emanating from out of sample non-manufacturing exporters. Models 5 and 6 examine the possibility of MMC occurring across battery and non-battery markets. Model 7 runs all the interactions in one specification. Coefficient standard errors are in brackets. $***p < .001$, $**p < .01$, $*p < .05$, $+p < .10$

10 Appendix: Collusive Outcomes from MMC

10.1 Competition in a Single Market

Suppose you have 2 infinitely rational firms bidding for contracts with a set of buyers in a single market. Each firm can either choose to play action C (cooperate) or D (defect). If both players cooperate, they evenly split Π^c, or the total welfare from colluding. If one player defects, they get Π^d or the profit from deviating. If both defect they operate at the competitive equilibrium $\Pi_i^N = max\{(c_j - c_i)Q_i, 0\}$, $j \neq i$ where c_i is the marginal cost of firm i and Q_i is the amount produced. Assume $\Pi^c \geq \Pi^d \geq \frac{\Pi^c}{2} \geq \Pi_i^N$ for $i = 1, 2$.[39] Table 1 shows a simple example of how competition between firms in a single market can be seen as a Prisoner's Dilemma game.

In a one shot game, the Nash equilibrium is for both firms to defect and operate at the competitive equilibrium, since each can always do better by deviating from cooperation. However, if these firms were to play this game across an infinite number of periods where they discount next periods payoff by $\delta \in [0, 1]$ with grim-trigger strategies, firm i will choose to cooperate if:

$$\frac{\Pi^c}{2(1-\delta)} \geq \Pi^d + \frac{\delta \Pi_i^N}{1-\delta} \tag{6}$$

From equation 6, it follows in equation 7:

$$\delta \geq \frac{2\Pi^d - \Pi^c}{2(\Pi^d - \Pi_i^N)} \tag{7}$$

Therefore if the benefit from deviating or firm i's cost advantage increases then the firm is more likely to choose defection (less of a likelihood that both firms are patient enough to cooperate.). On the other hand, an increase in the welfare from collusion would more likely lead to cooperation. If $\Pi^d = \Pi^c$ and $c_1 = c_2$, then 7 reduces to $\delta \geq \frac{1}{2}$, which is the result from firms competing in an infinitely repeated Bertrand game with grim-trigger strategies.

Let δ_i be the minimum discount rate firm i would need to cooperate.[40]. Again, assume $\Pi^d = \Pi^c$. If $c_1 > c_2$, then firm 1 will need to have $\delta_i \geq \frac{1}{2}$, but the lower bound for δ_2 will be lower than $\frac{1}{2}$. Put another way, if $\delta_1 = \delta_2 = \frac{1}{2}$ then firm 1 would choose to cooperate, but firm 2 will deviate and cooperation will not be sustained. Hence, asymmetry among firms in a single market leads to a lower likelihood of cooperation in this setting.

10.2 Competition in Multiple Markets

Now assume that firms 1 and 2 are competing in 2 markets A and B repeatedly over an infinite number of periods using grim-trigger strategies. Reference Table 2 to see how each market can be represented as its own Iterated Prisoner's Dilemma game.

[39] I make use of the asymmetric costs for firms when discussing how the game is played when the firms meet across multiple markets.

[40] So $\delta_i = \frac{2\Pi^d - \Pi^c}{2(\Pi^d - \Pi_i^N)}$

If both firms use unlinked strategies or operate in markets independently then equation 8 tells us that collusion in both markets can be sustained if:

$$\delta_i^{ind} \geq \max \left\{ \frac{2\Pi_k^d - \Pi_k^c}{2(\Pi_k^d - \Pi_{i,k}^N)} \right\}_{k=A,B} \tag{8}$$

for both $i = 1, 2$, where δ_i^{ind} is the patience needed for collusion to occur in both markets if firms operate independently across markets. However, if players use linked strategies or pool incentive constraints, then the separate games in Market A and Market B collapse into one game displayed in Table 3 below.

Unlike when firms treat markets as independent, when firms pool incentive constraints the action they choose must apply to both markets. That is, they cannot choose to cooperate in one market and defect in the other. Equation 9 shows the patience needed for firms to cooperate in both markets.

$$\delta_i^{pool} \geq \frac{2(\Pi_A^d + \Pi_B^d) - (\Pi_A^c + \Pi_B^c)}{2(\Pi_A^d + \Pi_B^d) - 2(\Pi_{i,A}^N + \Pi_{i,B}^N)} \tag{9}$$

where δ_i^{pool} is the lower bound for firm i's discount factor to cooperate if it is pooling incentive constraints across markets.

However, if firms have differing costs across markets, then collusion in both markets could be more likely. For simplicity assume that $\Pi_k^d = \Pi_k^c = \Pi$ in the analysis below.

Case 1 The single market outcome is replicated if firms have identical costs across markets (i.e. $\Pi_{i,A}^N = \Pi_{i,B}^N = 0$ for $i = 1, 2$) . That is, firm i must have $\delta_i^{pool} > \frac{1}{2}$.

Case 2: Assume $\Pi_{1A}^N > \Pi_{2A}^N$ and $\Pi_{1B}^N < \Pi_{2B}^N$. Then for firm 1:

$$\frac{2\Pi}{4\Pi - (\Pi_{1,A}^N + \Pi_{1,B}^N)} \leq \frac{\Pi}{2\Pi - \Pi_{1,A}^N)}$$

. The left hand side of the inequality is the threshold patience level when firm 1 is using linked strategies (δ_1^{pool}) and the right hand side is the threshold level for firm i in market A (δ_i^{ind}). Therefore, collusion is more likely in market A when the firm is pooling incentive constraints across markets, since there surely exists some level of discounting $\overline{\delta} \in [\delta_1^{pool}, \delta_1^{ind}]$ that will ensure collusion in market A when pooling incentive constraints. By symmetry, a similar argument can be made for firm 2 in market B.

Case 3: Assume $\Pi_{1A}^N > \Pi_{2A}^N$ and $\Pi_{1B}^N > \Pi_{2B}^N$. Without loss of generality, assume $\Pi_1 A^N \geq \Pi_1 B^N$. Then a similar argument can be made to Case 2. Using a pooled incentive constraint will still make collusion more likely. However, collusion in this instance is less likely compared to Case 2 when the firm only had sphere of influence in one market.

Therefore, with asymmetry across markets and firms. MMC can lead to a higher likelihood of collusion. However, symmetrical markets and firms would just replicate the single market outcome. This is the result obtained in Bernheim and Whinston (1990).